SMART INDIA HACKATHON • TECHNICAL ARCHITECTURE DOSSIER

AirTrace AI • Atmospheric Remote Sensing & Deep Learning PM2.5 Prediction Platform

Multi-Sensor Spaceborne Data Fusion, Hybrid CNN-BiLSTM-Attention Modeling & Physical Source-Receptor Transport Validation

Target Domain: Atmospheric AI & Remote Sensing

Sensors: Sentinel-5P TROPOMI • ERA5 • VIIRS/MODIS • INSAT-3D

Production Readiness: FastAPI Backend + Streamlit WebGL UI

1. Executive Summary & 30-Second Elevator Pitch

The Elevator Pitch:
Continuous ground-level Air Quality Monitoring Stations (CAAQMS) cover less than **0.01%** of India's landmass, leaving vast agricultural belts and industrial corridors completely unmonitored. **AirTrace AI** bridges this critical gap by fusing multi-sensor satellite remote sensing (Copernicus Sentinel-5P TROPOMI, MODIS/VIIRS FIRMS, MOSDAC INSAT-3D) with ECMWF ERA5 reanalysis and hybrid Deep Learning. The platform delivers:

1. **High-Precision Ground PM2.5 Spatial Prediction** anywhere in India via a dual **CNN-BiLSTM-Attention Neural Ensemble** (R^2 = **0.8899**, **MAE = 16.83 µg/m³**).
2. **Physical Source vs. Receptor Attribution** for reactive VOCs (Formaldehyde/HCHO) using **HDBSCAN Clustering** and **Physics-Informed Wind Dynamics** across 28 validated Indian domestic airsheds.

PREDICTION ACCURACY	MEAN ABSOLUTE ERROR	VALIDATED HOTSPOTS	INPUT TENSOR WIDTH
0.8899	**16.83**	**28**	**84**
R^2 Score on Test Data	µg/m³ (n=1,942 samples)	Indian Source Regions	Features × 14-Day Lags

2. End-to-End System Architecture Flowchart

LAYER 1: MULTI-SENSOR SPACEBORNE & METEOROLOGICAL INGESTION

Ingestion APIs

- **Sentinel-5P TROPOMI** — Daily Level-3 Rasters: CH4, CO, NO2, native HCHO (Terrascope openEO)
- **ECMWF ERA5 Reanalysis** — Hourly 10m U & V Wind Vectors, Boundary Layer Height (NetCDF4)
- **MODIS / VIIRS FIRMS** — Thermal Fire Radiative Power (FRP in MW) & Stubble Burning Coordinates
- **ISRO MOSDAC INSAT-3D** — Geostationary Aerosol Optical Depth (AOD) Diagnostics Adapter

⬇

LAYER 2: SPATIOTEMPORAL FEATURE ENGINEERING & SEQUENCE BUILDER

Tensor Preparation

- **Bounding Box Collocation** — Extracts target center ±0.05° training-compat spatial extent
- **14-Day Statistical Lags** — Computes mean, min, max, p10, p25, p75, p90, valid pixel fractions
- **Cyclical Solar Encodings** — sin/cos(DOY), sin/cos(Month), Coordinates, Days since origin
- **Missingness & Scaling** — MissingIndicator mask + RobustScaler → **Tensor (Batch, 14, 84)**

⬇

SUBSYSTEM A: DEEP LEARNING PM2.5 ENGINE

AI Inference

1. **1D-CNN Intra-Day Extraction** — Compresses sliding 3-day multi-gas chemical ratios & optical gradients
2. **Bidirectional LSTM Memory** — Processes forward & backward temporal context (14-day accumulation)

SUBSYSTEM B: HCHO & WIND OBSERVATORY

Transport Validation

1. **Spatial Candidate Anomaly Score**
   `S_i = [(C_i - μ_bg) / σ_bg] × [1 + ln(1 + FRP_i)]`
2. **HDBSCAN Density Clustering** — Filters background noise & isolates **28 Indian domestic source clusters**

3. Self-Attention Dynamic Pooling
Calculates attention weights α_t, focusing on flare & inversion spike days

4. Dual Regime Model Blending
`ŷ = 0.925 × ŷ_raw + 0.075 × ŷ_extreme` → **PM2.5 (ug/m3)**

3. Physics Wind Transport Dynamics
Evaluates Downwind Enhancement: `E_down = (C_down - C_up) / C_up`

4. Validation Confidence Tiers
Class A (Strong, p<0.05) | Class B (Moderate) | Class C (Diffuse)

⬇

💻 LAYER 4: FULL-STACK PRODUCTION & DECISION SUPPORT INTERFACE

Deployment

FastAPI Microservice API
Headless REST endpoints: `/predict`, `/health`, `/docs` (Uvicorn Port 8000)

Bento-Grid Streamlit UI
Interactive Mapbox layers, fly-to presets, color-coded NAAQS AQI gauges

WebGL GPU Acceleration
Hardware-accelerated point cloud explorer + sub-ms in-memory caching

NCAP Regulatory Matrix
Actionable airshed directives for CPCB / SPCB enforcement & CSV exports

3. Problem Statement & Technical Novelty

National Challenge	Traditional Approach / Industry Bottleneck	AirTrace AI Novel Solution
Spatial Data Void	Ground CAAQMS stations cost ~□1.5 Crore each; only major Tier-1 city centers are monitored.	**Zero-Hardware Satellite AI**: Maps ground concentrations across any coordinate/bounding box in India using Sentinel-5P Level-3 orbits.
Extreme Winter Episodic Spikes	Standard statistical regressions suffer from regression-to-the-mean, collapsing during severe smog events (>200 µg/m^3).	**Dual Regime Blending**: Fuses a Raw CNN-BiLSTM with an Extreme-Tail Neural Model (0.925 / 0.075 blend), eliminating extreme episode bias.
Source vs. Receptor Ambiguity	Regulators frequently penalize downstream districts that are merely receiving advected smoke plumes.	**Physical Wind Transport Engine**: Fuses ERA5 10m wind vectors with satellite columns to mathematically verify emission origin vs downwind advection ($p < 0.05$).
Precursor VOC Coupling	PM2.5 is modeled in isolation without accounting for Secondary Organic Aerosol (SOA) photochemical synthesis.	**HCHO-to-PM2.5 Synergy**: Uses tropospheric Formaldehyde as an observable satellite tracer for active NMVOC oxidation and SOA formation.

4. Deep Dive: Subsystem 1 — Dual CNN-BiLSTM-Attention PM2.5 Spatial AI

The prediction pipeline ingests a 14-day chronological sequence of spaceborne observations centered on the target bounding box. The 53 raw features are partitioned as follows:

- **Methane (CH4, 12 features)**: Spatial mean, median, std, min, max, p10, p25, p75, p90, valid pixel count, valid fraction, point value.
- **Carbon Monoxide (CO, 12 features)**: Tropospheric column statistical percentiles + data validity fraction (incomplete combustion tracer).
- **Nitrogen Dioxide (NO2, 12 features)**: Tropospheric vertical column distribution (vehicular and thermal combustion marker).
- **Formaldehyde (HCHO, 4 features)**: Native-resolution value, validity flag, pixel count, valid fraction (NMVOC oxidation proxy).
- **Spatiotemporal Coordinates (13 features)**: Latitude, Longitude, Year, Month, Day of Year, Week, Day of Week, Days Since Origin, Ordinal Date, `sin(DOY)`, `cos(DOY)`, `sin(Month)`, `cos(Month)`.

5. Deep Dive: Subsystem 2 — HCHO Observatory & Wind Transport Validation

Tier / Class	Validation Criteria	Sample Hotspot Clusters Identified	Actionable Regulatory Directive
Class A (Strong)	Persistence ≥ 0.70, Significant Downwind Enhancement ($p < 0.05$), High Composite Source Score.	**#34 Sangli-Satara Sugar Belt (MH)** **#12 Bharuch-Surat Petrochem (GJ)** **#26 Hoshangabad-Bhopal (MP)**	Mandatory Continuous Emission Monitoring (CEMS), Leak Detection & Repair (LDAR) audits, boiler retrofits.
Class B (Moderate)	Moderate persistence ($0.40 \le P < 0.70$), consistent seasonal transport plume.	**#1 Ludhiana-Jalandhar Belt (PB)** **#16 Bilaspur-Korba Power Basin (CG)** **#4 Coimbatore-Tiruppur (TN)**	In-situ crop residue management (CRM), Flue Gas Desulfurization (FGD), coal conveyor dust suppression.
Class C (Diffuse)	Ephemeral seasonal agricultural burning or diffuse regional background.	**#0 Bathinda-Sirsa Stubble Belt** **#24 Gorakhpur-Deoria Terai (UP)**	Seasonal satellite surveillance, biomass pelletization subsidies, farm-level advisory alerts.

6. Quantitative Model Benchmarks & Validation Results

Overall Test Set Performance (n=1,942)

Extreme Episodic Tail Breakdown

Evaluation Metric	AirTrace AI	Standard LSTM
R^2 (Variance Explained)	**0.8899**	0.7140
MAE (Mean Absolute Error)	**16.83 µg/m³**	26.40 µg/m³
RMSE (Root Mean Squared Error)	**26.43 µg/m³**	39.80 µg/m³
Median Absolute Error	**11.23 µg/m³**	18.50 µg/m³

Actual PM2.5 Regime	Samples (n)	Model MAE	Prediction Bias
All Days (> 0 µg/m³)	1,942	16.83 µg/m³	+0.35 µg/m³
Moderate (> 100 µg/m³)	633	25.71 µg/m³	-2.75 µg/m³
Severe (> 200 µg/m³)	239	29.83 µg/m³	-7.82 µg/m³
Hazardous (> 400 µg/m³)	7	41.52 µg/m³	**-0.19 µg/m³**

7. Defensible Answers to Key Judge Inquiries

Q1: "Satellite sensors measure total vertical column density in the stratosphere/troposphere. How can you reliably infer ground-level respirable PM2.5?"

Answer: Column density alone cannot determine surface concentrations. That is why our architecture does not perform naive linear regression. We engineer a 14-day temporal lag across 4 distinct chemical species (CH4, CO, NO2, HCHO) combined with cyclical solar/seasonal encodings and coordinates. The 1D-CNN extracts multi-gas optical depth ratios, while the BiLSTM-Attention captures planetary boundary layer (PBL) height collapse and surface inversion trapping over time, mapping optical columns into surface PM2.5 with **R^2 = 0.8899**.

Q2: "What happens when satellite data is missing due to cloud cover or during monsoon months?"

Answer: We built an automated data quality pipeline. The extractor computes `valid_count` and `valid_fraction` for each raster pass. The model input layer incorporates a scikit-learn `MissingIndicator` that flags missing channels, followed by a fitted `SimpleImputer`. Furthermore, if fewer than 7 days in the 14-day input sequence contain valid satellite observations, the system explicitly returns an 'insufficient satellite history' advisory rather than generating hallucinated estimates.

Q3: "How does your HCHO Source-Receptor Transport model help State Pollution Control Boards (SPCBs) under NCAP?"

Answer: Historically, regulatory bodies penalize receptor regions for elevated pollution without proving emission origin. By vector-multiplying ERA5 wind dynamics with Sentinel-5P columns and MODIS thermal fire radiative power, AirTrace AI calculates downwind plume enhancements with statistical p-values. This gives regulators legally defensible evidence to differentiate between local industrial polluters and incoming trans-boundary smoke plumes.

Q4: "Is this system scalable and production ready?"

Answer: Yes. The system is architected as a fully decoupled production platform: a high-throughput **FastAPI REST API** (exposing `/predict`, `/health`, `/model-contract`) for machine-to-machine integrations, and a responsive **Bento-Grid Streamlit SaaS Dashboard** utilizing WebGL hardware-accelerated rendering and sub-millisecond in-memory caching.

8. 3-Minute Live Presentation & Pitch Script

0:00 – 0:45	**The Problem & Dashboard Overview:** Introduce the 0.01% CAAQMS ground sensor coverage gap in India. Show the **Bento-Grid Dashboard** overview with 28 validated Indian domestic source regions and monthly fire dynamics.
0:45 – 1:45	**Subsystem 1 Live Inference (PM2.5 AI Engine):** Select *Delhi Anand Vihar* or *Ludhiana Industrial* preset. Run prediction to show the instant PM2.5 calculation (185 µg/m^3), color-coded NAAQS AQI classification, raw vs extreme model blend, and the 14-day multi-sensor satellite timeline.
1:45 – 2:30	**Subsystem 2 HCHO Wind Transport:** Fly to the *Western Petrochem Corridor (Bharuch-Surat)* on the interactive map. Open **Transport & Dynamics** to demonstrate the downwind plume enhancement curve against upwind baselines and ERA5 wind speed.
2:30 – 3:00	**Impact & NCAP Action Matrix:** Present the **NCAP Action Matrix** with targeted directives for SPCBs. Close with our key value proposition: *Zero-hardware cost, nationwide scalability, and verifiable mathematical accuracy.*

9. Production Execution Reference

1. Visual Web Dashboard

```
streamlit run unified_dashboard.py
```

Interactive UI on http://localhost:8501

2. FastAPI Backend API

```
uvicorn app.pipeline:app --host 0.0.0.0 --port 8000
```

REST API & Swagger Docs on http://localhost:8000/docs